Cut Locus of Submanifolds:
Stability and Intersection with Focal Locus

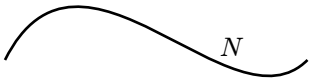
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IIT Delhi
30th October, 2026

Introduction

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$$p := \gamma(0) \in N, \quad \gamma(L) = q, \quad \mathrm{Len}(\gamma) = L = \mathrm{dist}_{\mathfrak{g}}(N, q).$$



Definition

The cut locus $\mathrm{Cut}(N)$ of a submanifold $N \subset M$ is the set of points $q \in M$ such that there exists an N-segment joining N to q, any extension of which fails to be distance-minimizing from N.

$$M \setminus \mathrm{Cut}(N) \underset{\text{s.d.r}}{\simeq} N, \qquad M \setminus N \underset{\text{s.d.r}}{\simeq} \mathrm{Cut}(N).$$

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Given $N \subset M$, the separating set of N is the collection

$$\mathrm{Sep}(N) := \{q \in M \mid \text{There exist at least two distinct } N\text{-segments joining } N \text{ to } q\}$$

Theorem : (_, Prasad [2024][†])

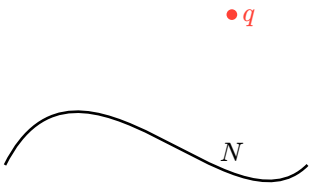
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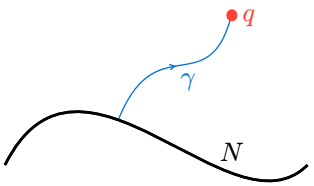
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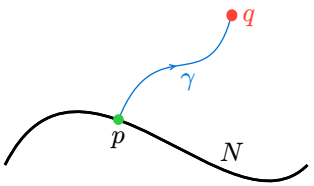
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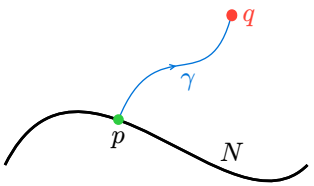
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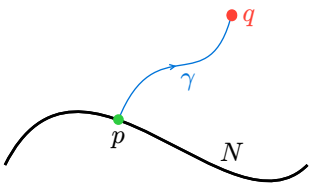
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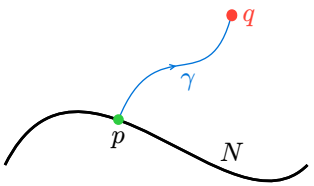
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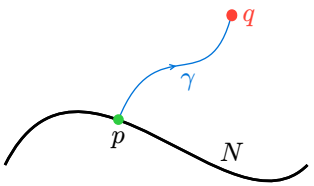
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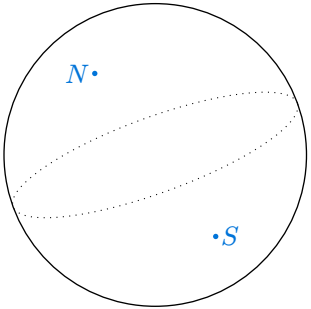
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Examples

The cut locus of the south pole S on the sphere S^2 consists of the north pole N.



More generally, the cut locus of a point $p \in S^n$ is the singleton $\{-p\}$

The cut locus of the equatorial S^1 in the sphere S^2 is the set $\{N, S\}$, which is S^0.

More generally, the cut locus of $S^k \subset S^n$ is S^{n-k-1}

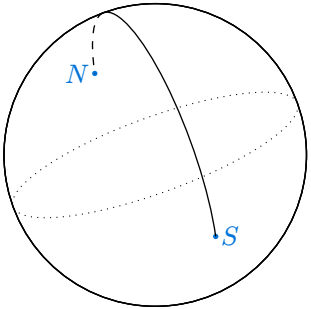
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The cut locus of $O(n, \mathbb{R})$ in $M(n, \mathbb{R})$ is $M(n, \mathbb{R}) \setminus \mathrm{GL}(n, \mathbb{R})$, the set of singular matrices (Basu-Prasad [2023][†]). This gives a geometric interpretation of the Gram-Schmidt process.

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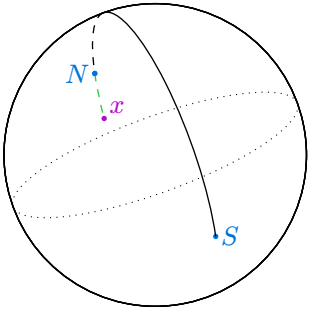
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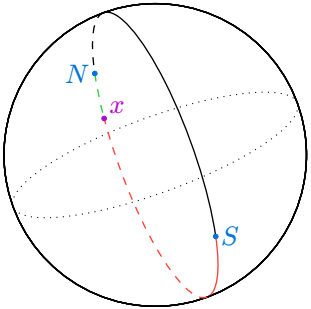
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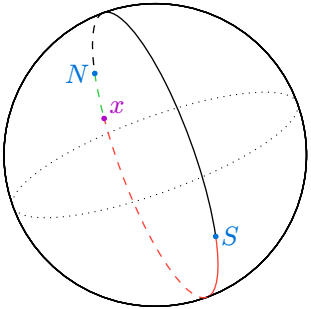
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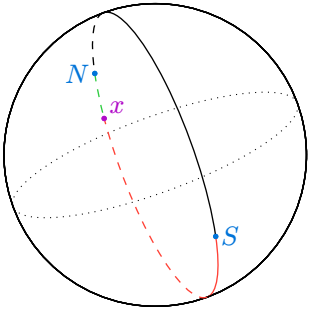
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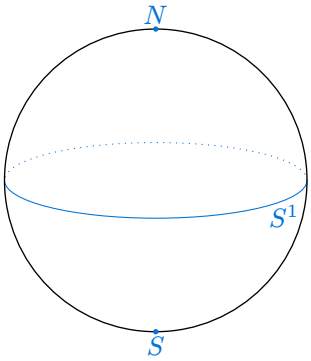
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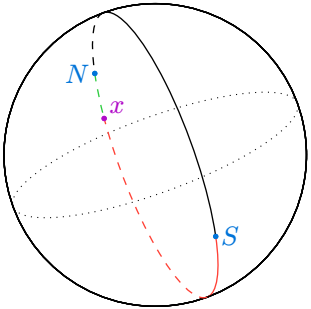
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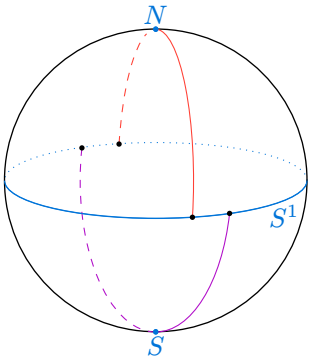
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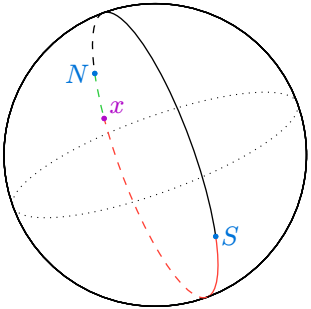
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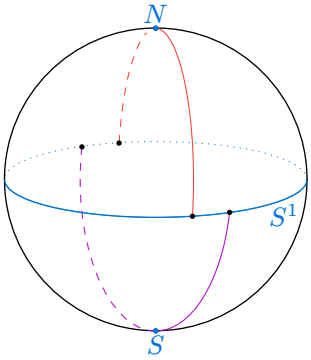
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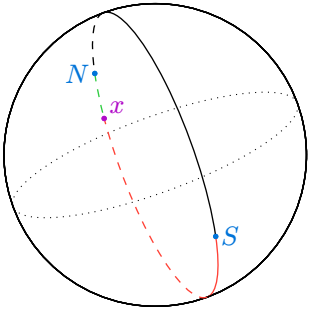
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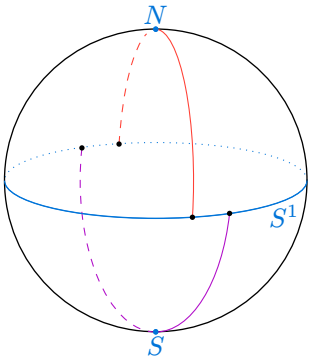
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Examples

The cut locus of the south pole S on the sphere S^2 consists of the north pole N.

More generally, the cut locus of a point $p \in S^n$ is the singleton $\{-p\}$





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More generally, the cut locus of $S^k \subset S^n$ is S^{n-k-1}

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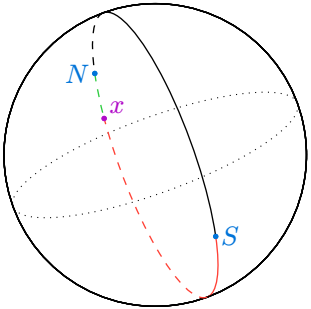
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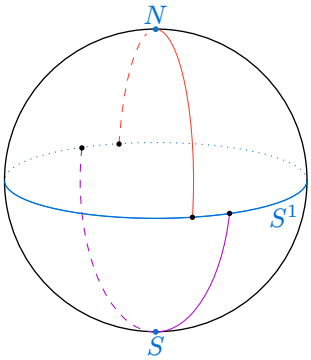
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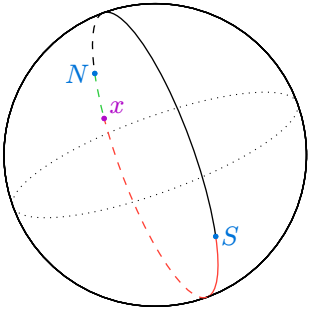
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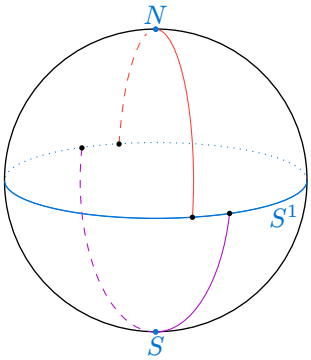
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Cut Time and Injectivity Radius

Given a unit normal vector $\mathbf{n} \in TN^{\perp_{\mathfrak{g}}}$, the cut time is defined as

$$\rho_{\mathfrak{g}}(N, \mathbf{n}) := \sup\left\{t \mid \operatorname{dist}_{\mathfrak{g}}(N, \gamma^{\mathfrak{g}}_{\mathbf{n}}(t)) = t\right\},$$

where $\gamma^{\mathfrak{g}}_{\mathbf{n}}$ is the unique geodesic w.r.t. $\mathfrak{g}$, with initial velocity $\mathbf{n}$.

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Suppose $\mathfrak{g}_i$ is a sequence of Riemannian metrics on a closed manifold M, converging to the metric $\mathfrak{g}_0$ in the Whitney C^2 topology. Then, $\lim_i \operatorname{Inj}(\mathfrak{g}_i) = \operatorname{Inj}(\mathfrak{g}_0)$. In other words, $\mathfrak{g} \mapsto \operatorname{Inj}(\mathfrak{g})$ is continuous.

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- **Buchner (1977)[†]:** $\mathrm{Cut}(p, \mathfrak{g})$ is stable from the point of view of singularity theory for generic $\mathfrak{g}$, on a manifold of dimension ≤ 6.
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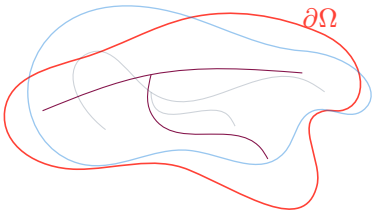
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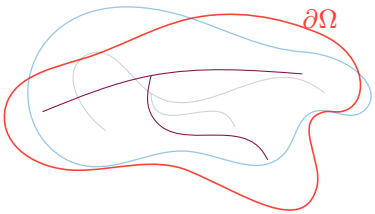
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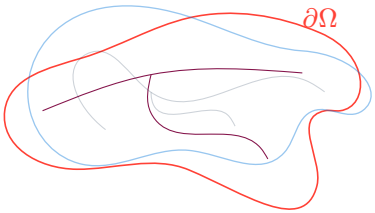
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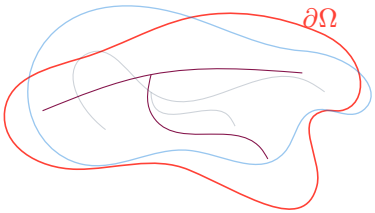
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The stability of the cut locus is well-studied in the literature from different perspectives.

- **Buchner (1977)[†]:** $\mathrm{Cut}(p, \mathfrak{g})$ is stable from the point of view of singularity theory for generic $\mathfrak{g}$, on a manifold of dimension ≤ 6.



Cut locus of $\partial\Omega$ inside Ω

- **Ardoy-Guijarro (2011)[‡]:** For an open set $\Omega \subset M$ with boundary $\partial\Omega$, $\mathrm{Sep}(\partial\Omega, \mathfrak{g}) \cap \Omega$ is Hausdorff stable when the boundary $\partial\Omega$ is perturbed by C^2-small diffeomorphisms.
- **Albano (2016)[§]:** $\mathrm{Cut}(\partial\Omega, \mathfrak{g})$ is Hausdorff stable, when both the boundary $\partial\Omega$ of a bounded domain $\Omega \subset \mathbb{R}^n$, and the metric are perturbed in a C^2-small sense.

- **Eltzner-García-Huckemann-Tuschmann (2021)[⁕]:** Cut locus is topologically stable from the point of view of metric measure theory.

In recent work, we have adapted the approach of Albano [2016][§].

[†] Michael A. Buchner. *Stability of the cut locus in dimensions less than or equal to* 6. `Invent. Math.` (1977).

[‡] Pablo Angulo Ardoy and Luis Guijarro. *Cut and singular loci up to codimension* 3. `Ann. Inst. Fourier (Grenoble).` (2011).

[§] Paolo Albano. *On the stability of the cut locus.* `Nonlinear Anal.` (2016).

[⁕] Benjamin Eltzner, Fernando Galaz-García, Stephan Huckemann, and Wilderich Tuschmann. *Stability of the cut locus and a central limit theorem for Frechet means of Riemannian manifolds.* `Proc. Amer. Math. Soc.` (2021).

Cut Locus as a Map

Given closed sets $A, B \subset M$ of a metric space (M, d), the Hausdorff distance between them is

$$d_H(A, B) := \max\left\{\sup_{a\in A} \operatorname{dist}(a, B), \quad \sup_{b\in B} \operatorname{dist}(A, b)\right\}.$$

Let M be a compact manifold without boundary, and $\mathfrak{h}$ be an auxiliary metric on M.

- Let $\mathcal{G} = \mathcal{G}(M)$ be the collection of C^2 Riemannian metrics on M, equipped with the Whitney C^2-topology.
- Let $\mathcal{K} = \mathcal{K}(M) := \{K \subset M \mid K \text{ is compact}\}$, equipped with the topology induced by the Hausdorff metric d_H associated with $\mathfrak{h}$. The topology is independent of the choice of $\mathfrak{h}$.
- Given a manifold N, let $\operatorname{Emb}(N, M)$ be the space of smooth embeddings $N \hookrightarrow M$, equipped with the Whitney C^2-topology.

As the cut locus is closed, we have the maps

$$\begin{aligned} \operatorname{Cut} : M \times \mathcal{G} &\to \mathcal{K} \\ (p, \mathfrak{g}) &\mapsto \operatorname{Cut}(p, \mathfrak{g}), \end{aligned} \qquad \begin{aligned} \operatorname{Cut} : \operatorname{Emb}(N, M) \times \mathcal{G} &\to \mathcal{K} \\ (f, \mathfrak{g}) &\mapsto \operatorname{Cut}(f(N), \mathfrak{g}), \end{aligned}$$

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Stability of Cut Locus and Injectivity Radius

Theorem : (_, Itoh, Prasad [2026][†])

The cut locus map $\mathrm{Cut} : M \times \mathcal{G} \to \mathcal{K}$ is continuous.

Recently, we have generalized the result of Sakai [1983].

Theorem : (_ [2026][‡])

Given closed manifolds N, M, the injectivity radius map

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As a consequence, we get the following.

Theorem : (_ [2026][‡])

Given a closed manifold N, the cut locus map $\mathrm{Cut} : \mathrm{Emb}(N, M) \times \mathcal{G} \to \mathcal{K}$ is continuous.

As the spaces involved are metrizable, we only need to verify sequential continuity.

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Proof Sketch of Stability: Ingredients

We have the characterization of convergence in the Hausdorff metric.

Lemma

Suppose M is compact, $X_i, X \subset M$ are closed (and hence compact) sets. Then, $\lim_i d_H(X_i, X) = 0$ if and only if the following holds:

- for any $x \in X$, there exists $x_i \in X_i$ with $\lim_i x_i = x$, and
- for any convergent sequence $x_{i_j} \in X_{i_j}$, we have $\lim_j x_{i_j} \in X$.

We also have a useful characterization of a separating point. Given $\Omega \subset M$ open, consider the cut locus (resp. separating set) of the boundary $\partial\Omega$ inside Ω:

$$\mathrm{Cut}^{\mathrm{in}}(\Omega, \mathfrak{g}) := \mathrm{Cut}(\partial\Omega, \mathfrak{g}) \cap \Omega, \qquad \mathrm{Sep}^{\mathrm{in}}(\Omega, \mathfrak{g}) := \mathrm{Sep}(\partial\Omega, \mathfrak{g}) \cap \Omega.$$

The distance function $u := \mathrm{dist}_{\mathfrak{g}}(\partial\Omega, _)$ is *not* C^1 on points of $\mathrm{Sep}^{\mathrm{in}}(\Omega, \mathfrak{g})$, and the C^1 singular support of u inside Ω is precisely $\mathrm{Cut}^{\mathrm{in}}(\Omega, \mathfrak{g})$.

Lemma

$x \in \mathrm{Sep}^{\mathrm{in}}(\Omega, \mathfrak{g})$ if and only if there exists a C^1 function φ near x such that $u - \varphi$ attains a (strict) local maximum at x and $\|d_x\varphi\|_{\mathfrak{g}} < 1$.

The characterization is a consequence of the fact that u is the unique (positive) viscosity solution of the eikonal equation $\|\nabla f\|_{\mathfrak{g}} = 1$ with $f|_{\partial\Omega} = 0$.

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Suppose $\mathfrak{g}_i \to \mathfrak{g}_0$ is a C^2-convergent sequence of metrics, and $f_i \to f_0$ is a C^2-convergent sequence of embeddings $N \hookrightarrow M$; set $N_i = f_i(N)$. Let $\mathbf{n}_i \in TN_i^{\perp^{\mathfrak{g}_i}}$ be unit normal vectors converging to some $\mathbf{n}_0$, necessarily in $TN_0^{\perp^{\mathfrak{g}_0}}$ with $\|\mathbf{n}_0\|_{\mathfrak{g}_0} = 1$. Then, $\lim_i \rho_{\mathfrak{g}_i}(N_i, \mathbf{n}_i) = \rho_{\mathfrak{g}_0}(N_0, \mathbf{n}_0)$.

The above strengthens the classical result about the continuity of the cut time map of a fixed submanifold, with respect to a fixed metric.

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Focal Points

A cut point may not always be a separating point.

Let N be an ellipse in $\mathbb{R}^2$, with focal points e, e'.

$$\mathrm{Sep}(N) = \textit{Open} \text{ interval between } e \text{ and } e'.$$

For some x on the open interval, we have $[p_1, x]$ and $[p_2, x]$ as N-segments. For e (resp. e'), we have *unique* N-segment $[a, e]$ (resp. $[b, e']$), any extension of the of which fails to be distance minimizing from N. Thus,

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Definition

Given $N \hookrightarrow M$, a point $q \in M$ is called a focal point of N if it is a *critical value* of the normal exponential map $\exp^{\perp} : TN^{\perp} \to M$

Theorem : (_, Prasad [2024][1])

For any $q \in \mathrm{Cut}(N)$, we have a *non-exclusive* dichotomy:

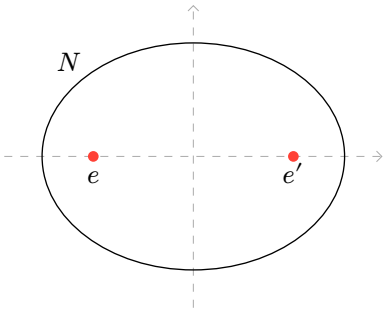
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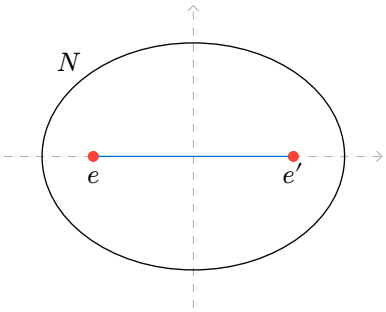
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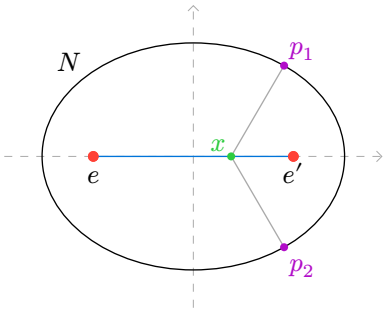
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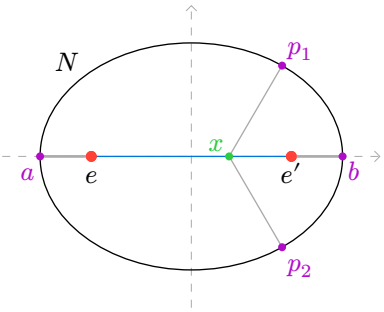
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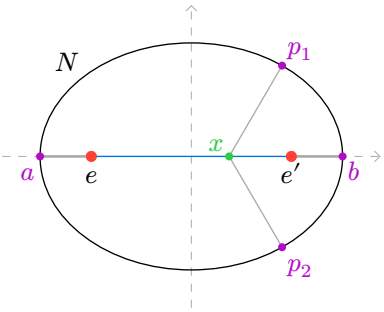
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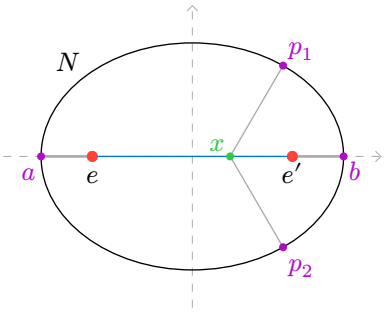
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Focal Points

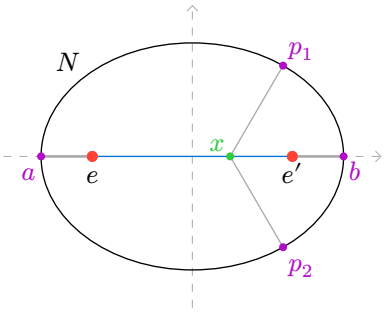
A cut point may not always be a separating point.

Let N be an ellipse in $\mathbb{R}^2$, with focal points e, e'.

$$\mathrm{Sep}(N) = \textit{Open} \text{ interval between } e \text{ and } e'.$$

For some x on the open interval, we have $[p_1, x]$ and $[p_2, x]$ as N-segments. For e (resp. e'), we have *unique* N-segment $[a, e]$ (resp. $[b, e']$), any extension of the of which fails to be distance minimizing from N. Thus,

$$\mathrm{Cut}(N) = \textit{Closed} \text{ interval joining } e \text{ and } e'.$$



Definition

Given $N \hookrightarrow M$, a point $q \in M$ is called a focal point of N if it is a *critical value* of the normal exponential map $\exp^{\perp} : TN^{\perp} \to M$

Theorem : (_, Prasad [2024][†])

For any $q \in \mathrm{Cut}(N)$, we have a *non-exclusive* dichotomy:

$q \in \mathrm{Sep}(N)$, or, q is the *first* focal point of N along some N-segment.

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Intersection of Cut and Focal Loci

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Conjecture : (Rauch [1959][†])

For any point p in a compact, simply connected, Riemannian manifold $(M, \mathfrak{g})$, the cut locus $\mathrm{Cut}(p, \mathfrak{g})$ must have a conjugate point of p along some distance-minimizing geodesic from p.

The conjecture is true for S^2, and for symmetric spaces.

Theorem : (Weinstein [1968][‡])

If M is a compact manifold without boundary, not homeomorphic to S^2, then there exists a metric $\mathfrak{g}$ on M and a point $p \in M$, such that the cut locus $\mathrm{Cut}(p, \mathfrak{g})$ has no conjugate point of p along any distance-minimizing geodesic from p.

We have generalized this to the case of submanifolds.

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An Estimate on the First Focal Time

Principal curvature : generalization of the curvature of a planar curve.

Sectional curvature : generalization of the Gaussian curvature of a surface.

Theorem : (Warner [1966][†])

Let $H \subset M$ be a codimension 1 submanifold in a compact Riemannian manifold $(M, \mathfrak{g})$, and $\mathbf{n}$ be a unit vector field normal to H at every point. Suppose $\Delta_H, \kappa_M > 0$ are real numbers such that

- the principal curvatures of H along $\mathbf{n}$ are $> -\Delta_H$, and
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Set

$$\varepsilon_H := \frac{1}{\sqrt{\kappa_M}} \arctan\left(\frac{\sqrt{\kappa_M}}{\Delta_H}\right).$$

Then, for any unit speed geodesic $\gamma : [0, \infty) \to M$ in the direction of $\mathbf{n}$, there is no focal point of H along γ in the interval $[0, \varepsilon_H]$.

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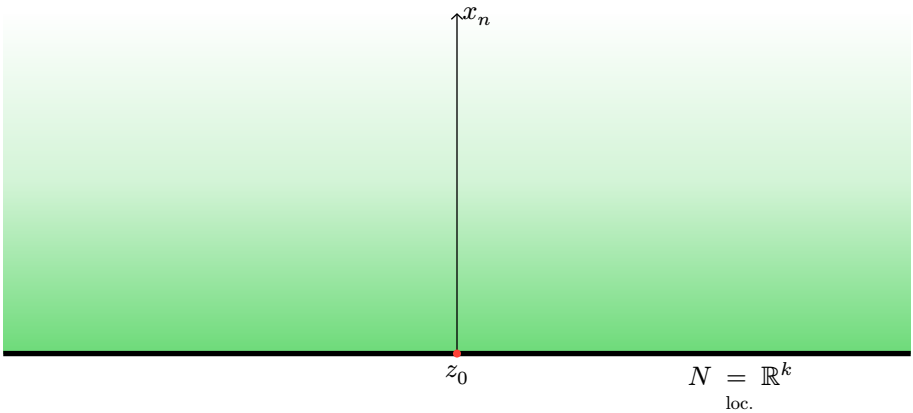
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Step 1: Choosing an Initial Metric and a Tubular Neighborhood

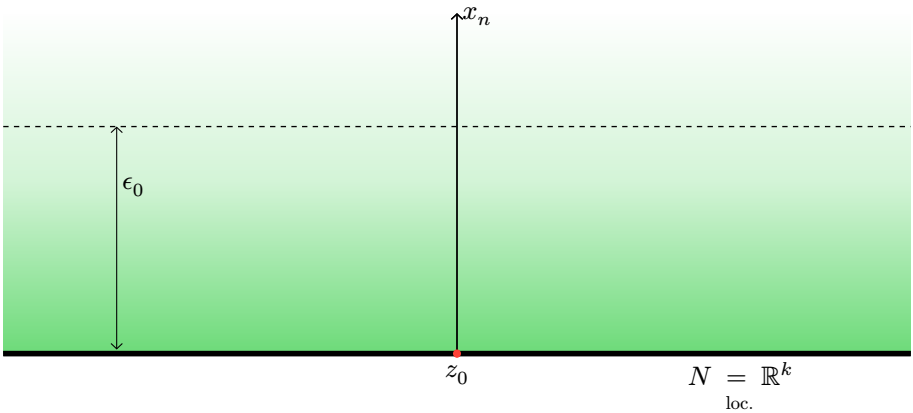
For $z_0 \in N$, get a metric which is Euclidean near z_0 and N is isometric to $\mathbb{R}^k \times \{0\} \subset \mathbb{R}^n$. Choose an ϵ_0-tubular neighborhood of N, which is $\mathbb{R}^k \times \mathbb{D}^{n-k}_{\epsilon_0}$ near z_0. Get a smaller tubular neighborhood Ω, whose boundary P is flat and parallel to $\mathbb{R}^{n-1} \times \{0\}$ near z_0.



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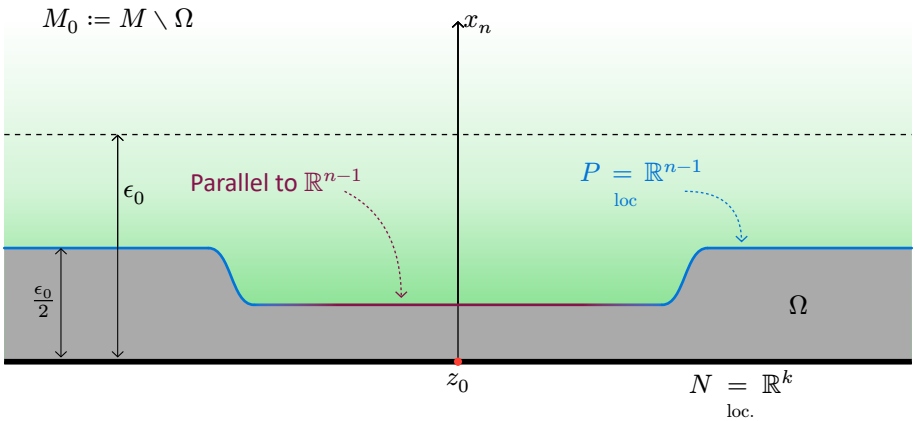
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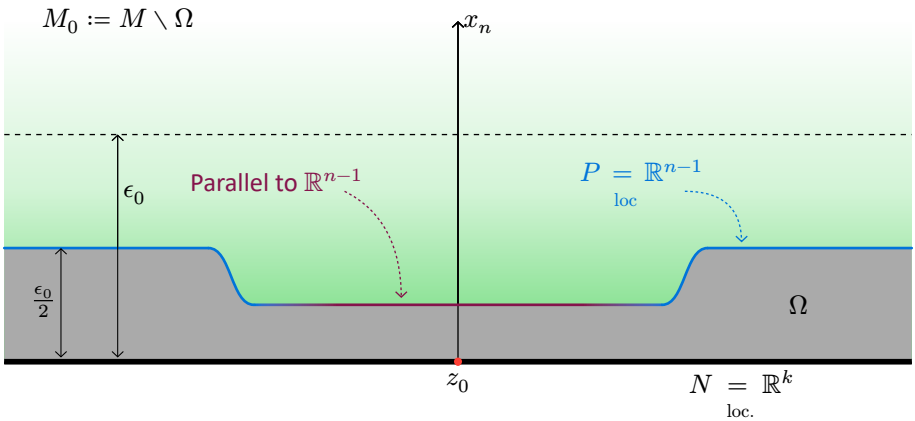
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Step 2: Constructing a Dense Curve with Bounded Curvature

Theorem : (_, Prasad, Schick)

There exists $\Delta_{\text{curve}} > 0$, such that we can get a regular curve in $M_0 \setminus P$, with geodesic curvature strictly bounded by Δ_{curve}, which is arbitrarily dense in a given compact set $\mathcal{K} \subset M_0 \setminus P$. The curve can be assumed to be embedded if $\dim M_0 \geq 3$.

Fix $\Delta_P, \kappa_M > 0$, such that principal curvatures of P are $> -\Delta_P$ along the outward normal directions, and sectional curvatures of M are $< \kappa_M$. Set,

$$\Delta_{\text{final}} := \max\left\{\sqrt{\frac{32}{27}}\Delta_{\text{curve}},\quad \Delta_P\right\}, \qquad \varepsilon_0 := \frac{1}{\sqrt{\kappa_M}}\arctan\left(\frac{\sqrt{\kappa_M}}{\Delta_{\text{final}}}\right).$$

Get ε_0-neighborhood of P, and set $\mathcal{K}$ as the region away from N, with boundary Q.

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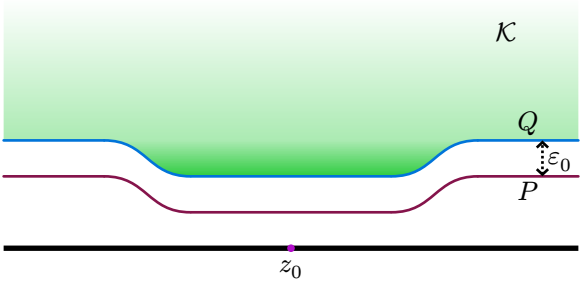
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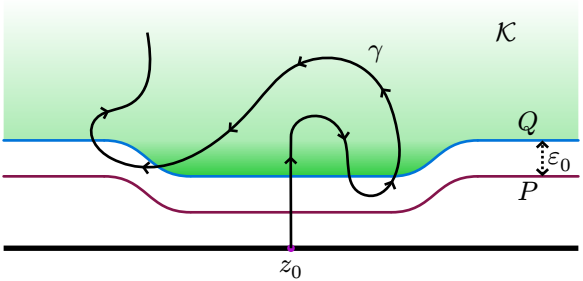
Theorem : (_, Prasad, Schick)

There exists $\Delta_{\text{curve}} > 0$, such that we can get a regular curve in $M_0 \setminus P$, with geodesic curvature strictly bounded by Δ_{curve}, which is arbitrarily dense in a given compact set $\mathcal{K} \subset M_0 \setminus P$. The curve can be assumed to be embedded if $\dim M_0 \geq 3$.

Fix $\Delta_P, \kappa_M > 0$, such that principal curvatures of P are $> -\Delta_P$ along the outward normal directions, and sectional curvatures of M are $< \kappa_M$. Set,

$$\Delta_{\text{final}} := \max\left\{\sqrt{\frac{32}{27}}\Delta_{\text{curve}},\quad \Delta_P\right\}, \qquad \varepsilon_0 := \frac{1}{\sqrt{\kappa_M}}\arctan\left(\frac{\sqrt{\kappa_M}}{\Delta_{\text{final}}}\right).$$

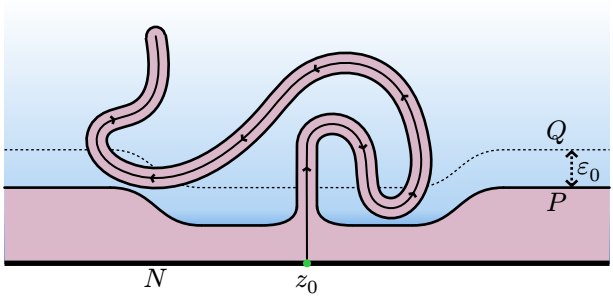
Get ε_0-neighborhood of P, and set $\mathcal{K}$ as the region away from N, with boundary Q.



Get embedded regular curve γ, $\frac{\varepsilon_0}{2}$-dense in $\mathcal{K}$, with strict geodesic curvature bound Δ_{curve}.

Step 3: Constructing a New Tubular Neighborhood

Next, we get a tube around the curve γ, and join the boundary smoothly to P.



The new region $\tilde{\Omega}$ is a tubular neighborhood of N, with boundary $\mathcal{B}$. We estimate the principal curvatures of $\mathcal{B}$.

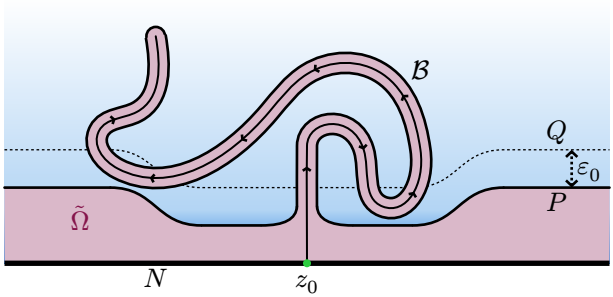
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- On P, the bound is $> -\Delta_P \geq -\Delta_{\text{final}}$.
- The funnel-shaped join is in a Euclidean region, where we can perform explicit computations.

By Warner's theorem: there are no focal points of $\mathcal{B}$ in the outward normal direction for time $[0, \varepsilon_0)$.

Hence, by the $\frac{\varepsilon_0}{2}$-denseness, a cut point of $\mathcal{B}$ on the outside $M_0 \setminus \tilde{\Omega}$ is *not* a focal point of $\mathcal{B}$ along any $\mathcal{B}$-segment.

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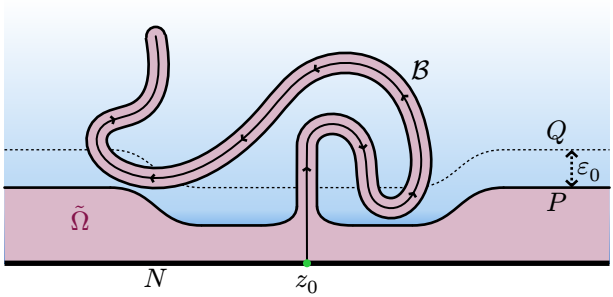
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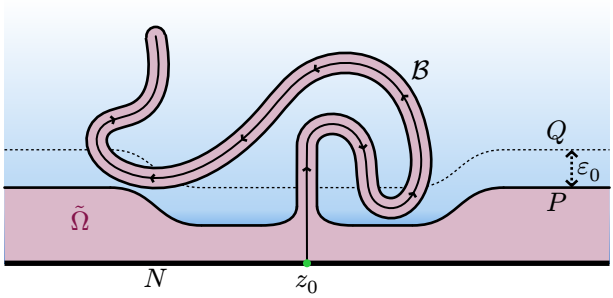
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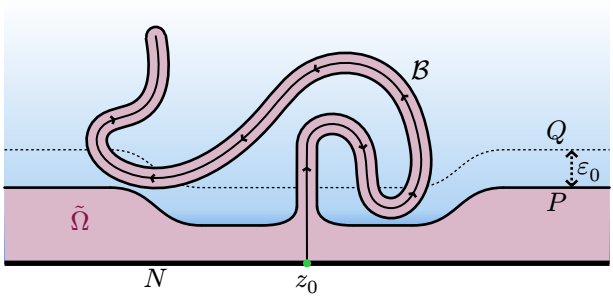
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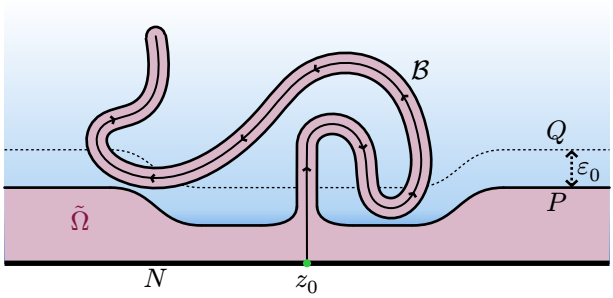
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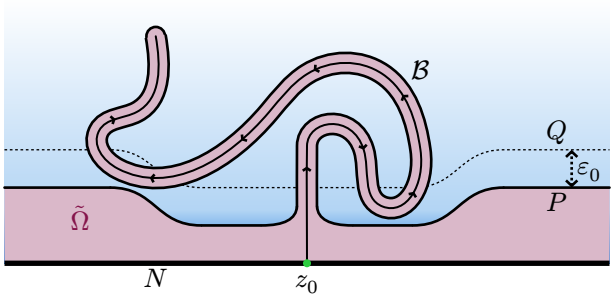
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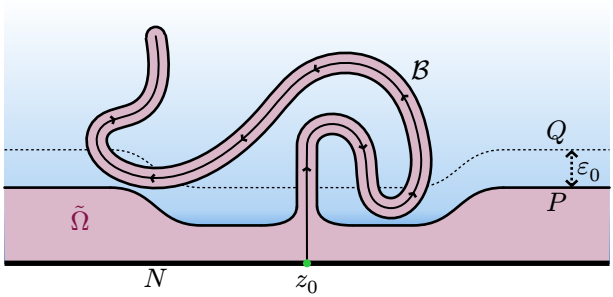
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Step 4: Changing the Metric inside the Tubular Neighborhood

Theorem : (_, Prasad, Schick)

There exists a new metric $\mathfrak{g}$ on M such that the following holds.

- $\mathfrak{g}$ is unchanged on the complement of the tubular neighborhood $\tilde{\Omega}$, near its boundary $\mathcal{B}$, and near the submanifold N.
- $\tilde{\Omega}$ is a *unit normal neighborhood* of N with respect to $\mathfrak{g}$.

Proof of Main Theorem : With the new metric $\mathfrak{g}$,

- every $\mathcal{B}$-segment $\gamma : [0, \epsilon] \to M_0 \setminus \tilde{\Omega}$ extends back to an N-segment $\tilde{\gamma} : [-1, \epsilon] \to M$, and
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Some Consequences

Theorem : (_, Prasad, Schick)

Let $N_1, N_2 \subset M$ be compact, connected, disjointly embedded, submanifolds, without boundary. Then, the following are equivalent.

- There is a metric $\mathfrak{g}$ on M such that $N_2 = \mathrm{Cut}(N_1, \mathfrak{g})$.
- M is diffeomorphic to the union of two disc bundles over N_1 and N_2, attached along the boundary by a diffeomorphism.
- There exists a Morse-Bott function with exactly two critical submanifolds N_1 and N_2.
- There exists a function $f : M \to \mathbb{R}$ with critical set $N_1 \sqcup N_2$.

Corollary : (B.-Prasad-Schick, Basu-Prasad [2026][†], Lerario-Meroni-Zuddas [2024][‡])

Given a closed manifold M, if there exists a function $f : M \to \mathbb{R}$ with exactly two critical values a, b, and if $S^p = f^{-1}(a), S^q = f^{-1}(b)$, with $p \neq q$, then M is homeomorphic to S^{p+q+1}.

Corollary : (_ [2026][§])

Let $(M, \mathfrak{h})$ be a closed Riemannian manifold. Suppose $N \subset M$ is a closed submanifold such that $\mathrm{Cut}(N, \mathfrak{h})$ is free of focal points of N along N-segments. Then, there exists a metric $\mathfrak{g}$ on M such that $\mathrm{Cut}(N, \mathfrak{g})$ is a finite subcomplex of M (and still free of focal points).

[‡] Somnath Basu and Sachchidanand Prasad. *On manifolds homeomorphic to spheres* (preprint) (2026).

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Ongoing Work

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Lorentzian geometry comes up in the study of space-time.

Both of them can be generalized under the framework of wind Finsler geometry.

Modeling Wildfire: Forrest wildfires are modeled via methods in Riemannian geometry; fire is concentrated at *focal points* of the front of the fire.

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Our immediate goals are the following.

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